Logi-Track
Group 1

SE 305/321 – Software Specification and Design
Term Project Design Document

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1. Introduction

Logi-Track is an end-to-end digital management and tracking platform specifically designed for the logistics sector, with a primary focus on sensitive and temperature-controlled container transportation. The system centralizes communication between operations teams, drivers, and customers into a single unified structure, aiming to eliminate fragmented manual workflows.
Operating through a web-based control panel for management and a dedicated mobile application for drivers, the project enables real-time monitoring of vehicle locations, container temperature levels, tire pressure, and axle status. By implementing Role-Based Access Control (RBAC) to ensure data security and integrating automated notifications with a manual pricing module, Logi-Track transforms traditional logistics operations into a transparent, agile, and highly secure process.

2.1. Problem Definition

Despite the global trend of digital transformation, the logistics industry still suffers from a heavy reliance on manual workflows, which creates significant operational bottlenecks. One of the most pressing issues is the inefficient flow of information; receiving bookings via email and managing updates through phone calls frequently leads to critical data loss, miscommunication, and costly delays. Furthermore, there is a distinct lack of real-time monitoring for perishable or temperature-sensitive goods. Without instantaneous data on interior climate and container integrity, companies face a high risk of cargo damage and financial loss. This inefficiency is compounded by a heavy manual workload, where tasks such as driver assignment and invoice generation are performed without automation, increasing the likelihood of human error and inflating operational costs. Ultimately, these factors culminate in visibility and trust issues, as customers lack direct access to the status of their shipments, resulting in constant inquiries and general dissatisfaction due to a lack of transparency.

2.2. Why We Wanted to Adress this Issue:

The primary motivation for addressing these challenges is to bridge the "information gap" in logistics operations by introducing modern technological standards that ensure reliability and scalability. By prioritizing operational transparency, we aim to eliminate uncertainty for both the service provider and the client by making every stage of the transportation process entirely data-driven and visible. Additionally, we are committed to enhancing safety and compliance; our goal is to minimize regulatory risks and product loss by guaranteeing that sensitive goods, such as pharmaceuticals and food, are transported under optimal and documented conditions. Through automation, we intend to replace burdensome manual workloads with streamlined systems, allowing operations teams to focus on strategic decision-making while increasing the overall profitability of the company. Ultimately, we believe that providing clients with a dedicated portal for traceability will build a foundation of trust and significantly elevate the quality of service in an increasingly competitive market.

3. Proposed System Design

3.1. Requirements

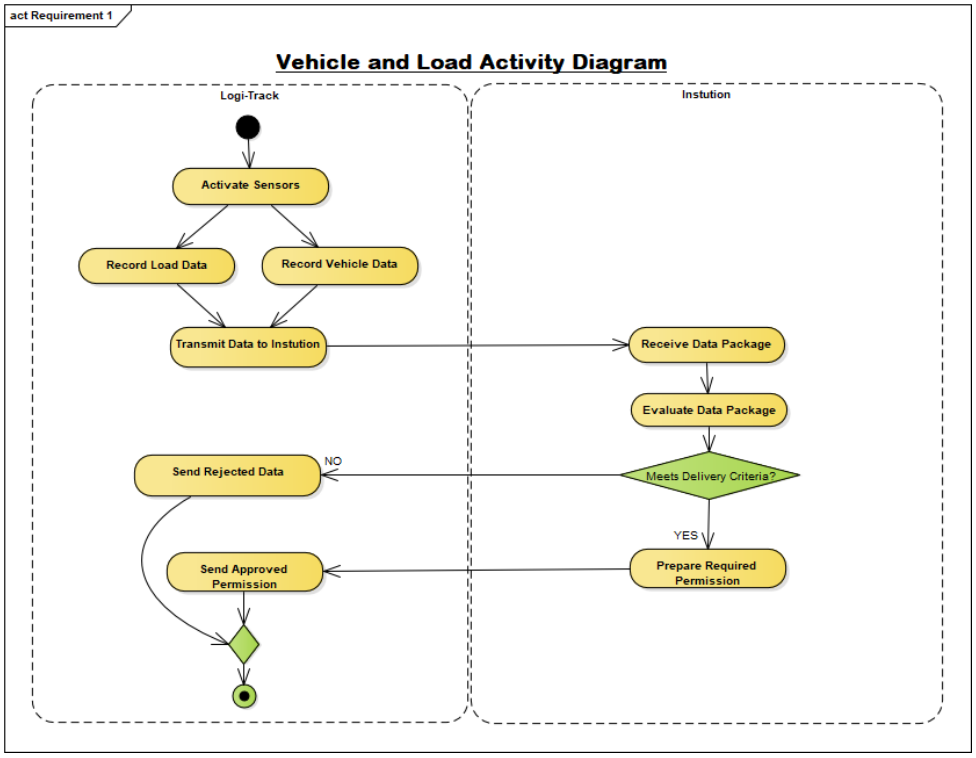
Non-Functional Requirements:

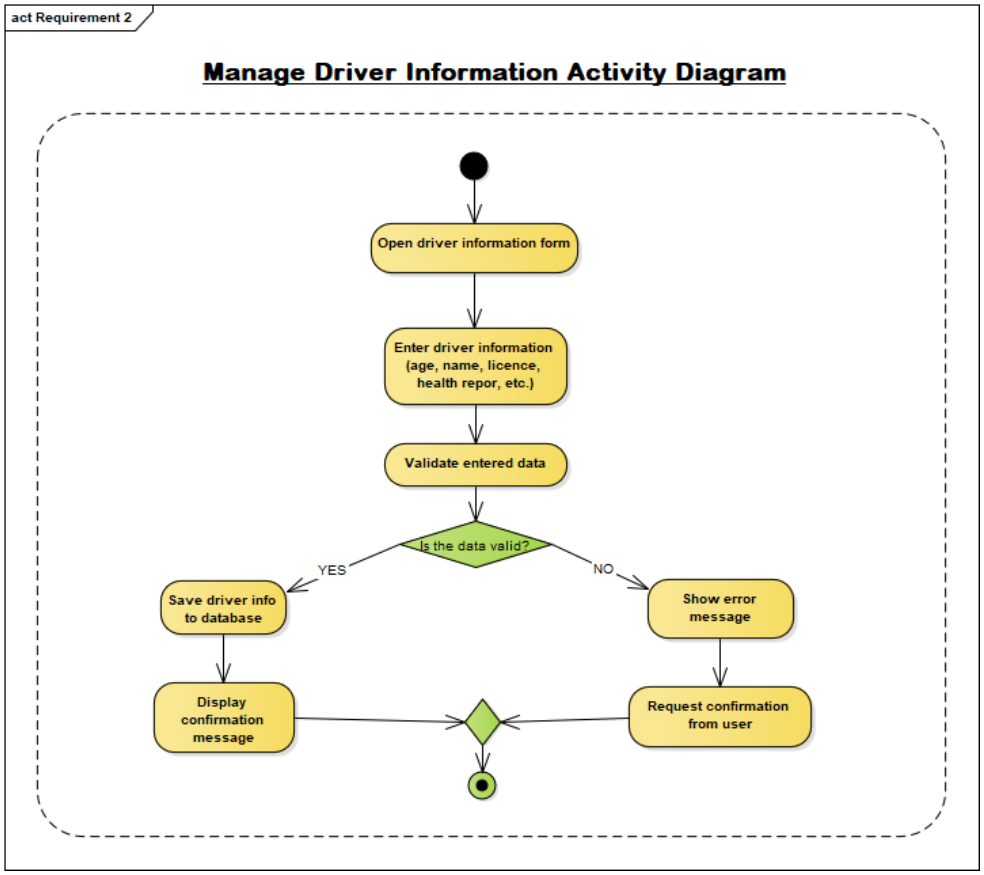
1- The app should allow first-time users to complete key actions (e.g., view a delivery, update a status) within 3 clicks and under 30 seconds.
2- Each user role (Admin, Operations Staff, Driver, Customer) will have a defined set of permissions restricting access to sensitive data.
3- Customers can only view high-level shipment status (*In Transit*, *Delivered*, *Delayed*) and estimated delivery time +-15 min accuracy.
4- Voice notification latency for route or status updates must be <=2 seconds after event trigger.
5- The system must automatically alert the driver after 4.5 hours of continuous driving to take a minimum 45-minute break.

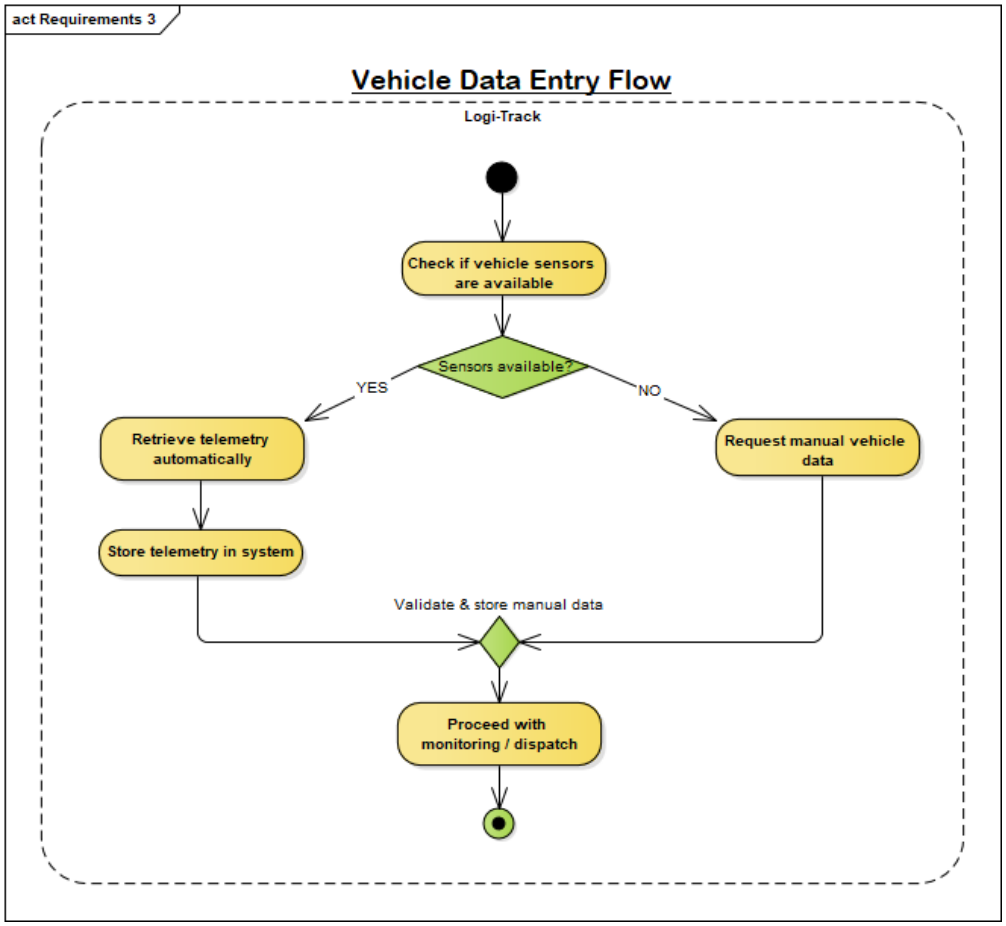
Functional Requirements:

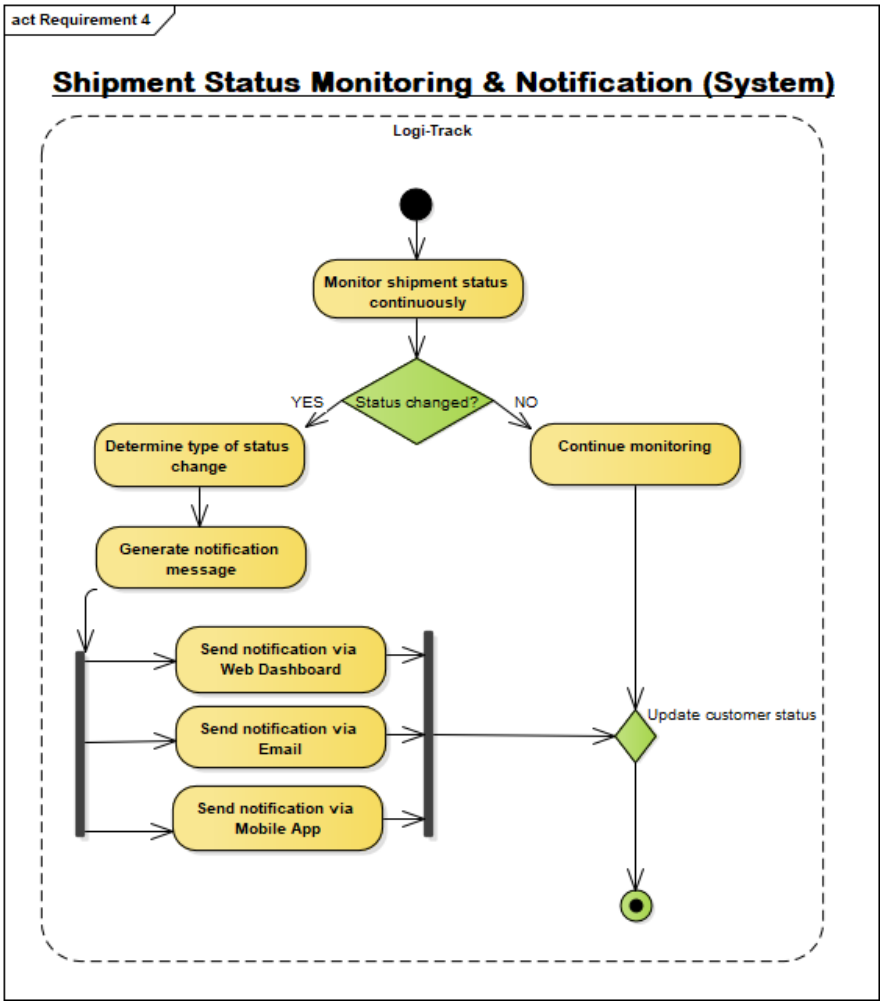
1- The system will record vehicle and load data using sensors and transmit it to the relevant institution.
2- It will contain detailed driver information (name, age, driver's license, health report, etc.).
3- The system will also allow manual entry of vehicle information, as some vehicles may not have sensors. In cases where vehicle data cannot be automatically retrieved from sensors, users must enter this information manually.
4- The system shall send notifications to the operations team when a shipment status changes.
5- When the driver confirms the delivery, the system will create an invoice record and send a notification to the finance department.
6- The application will include a tutorial section.
7- Newly assigned tasks will be notified to drivers via the mobile app.
8- Vehicle locations will be tracked via GPS.
9- Photo uploads will be available (e.g., in the event of container damage).
10- Drivers' work and rest times will be tracked.
11- Users will log in to the system with their own username and password.
12- All transactions will be recorded with a timestamp. Past transactions can be easily audited.

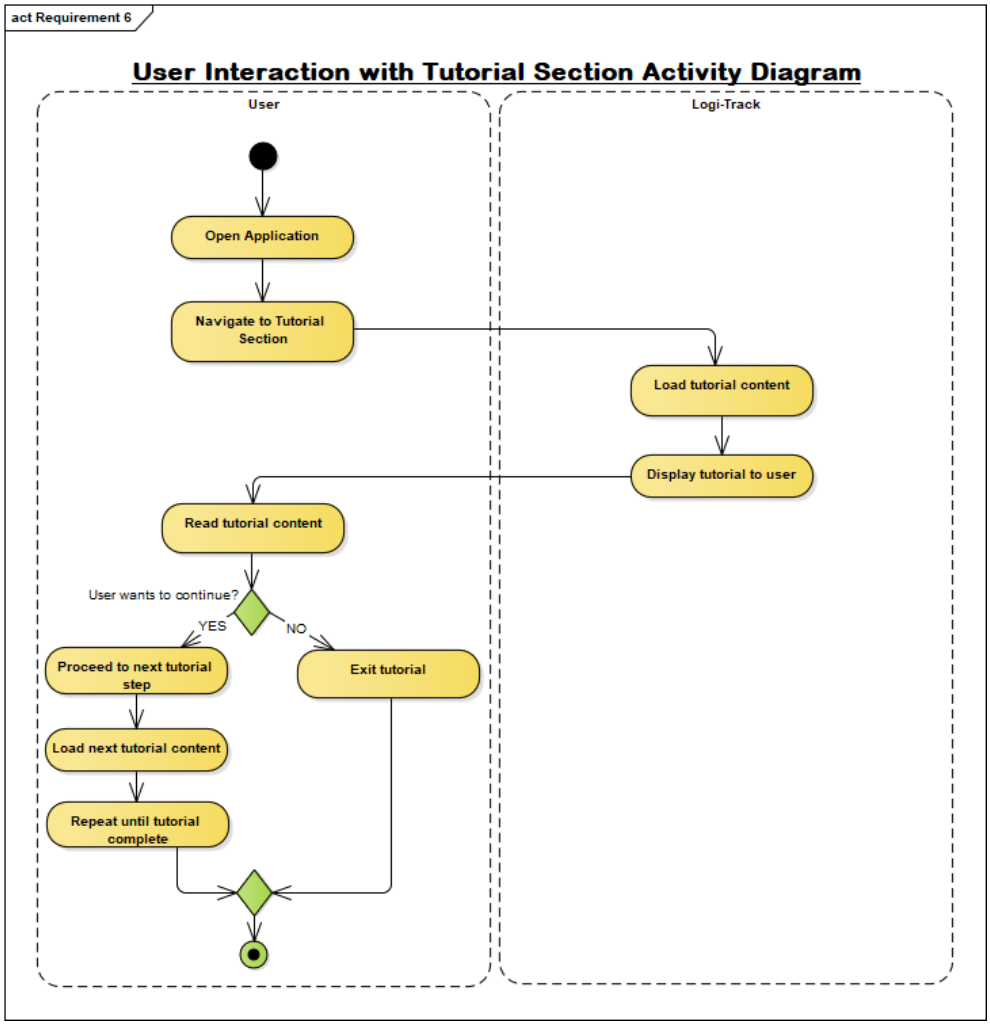
3.2. Activity Diagrams:

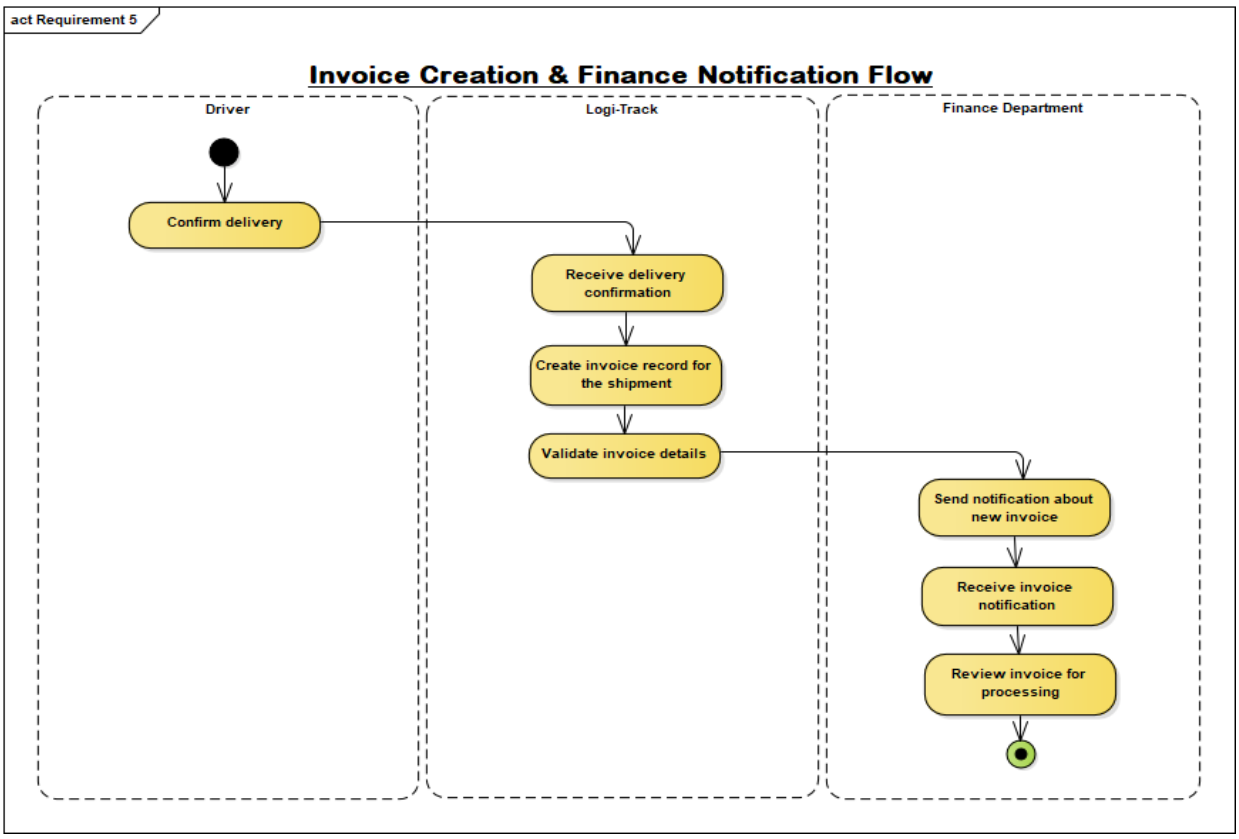


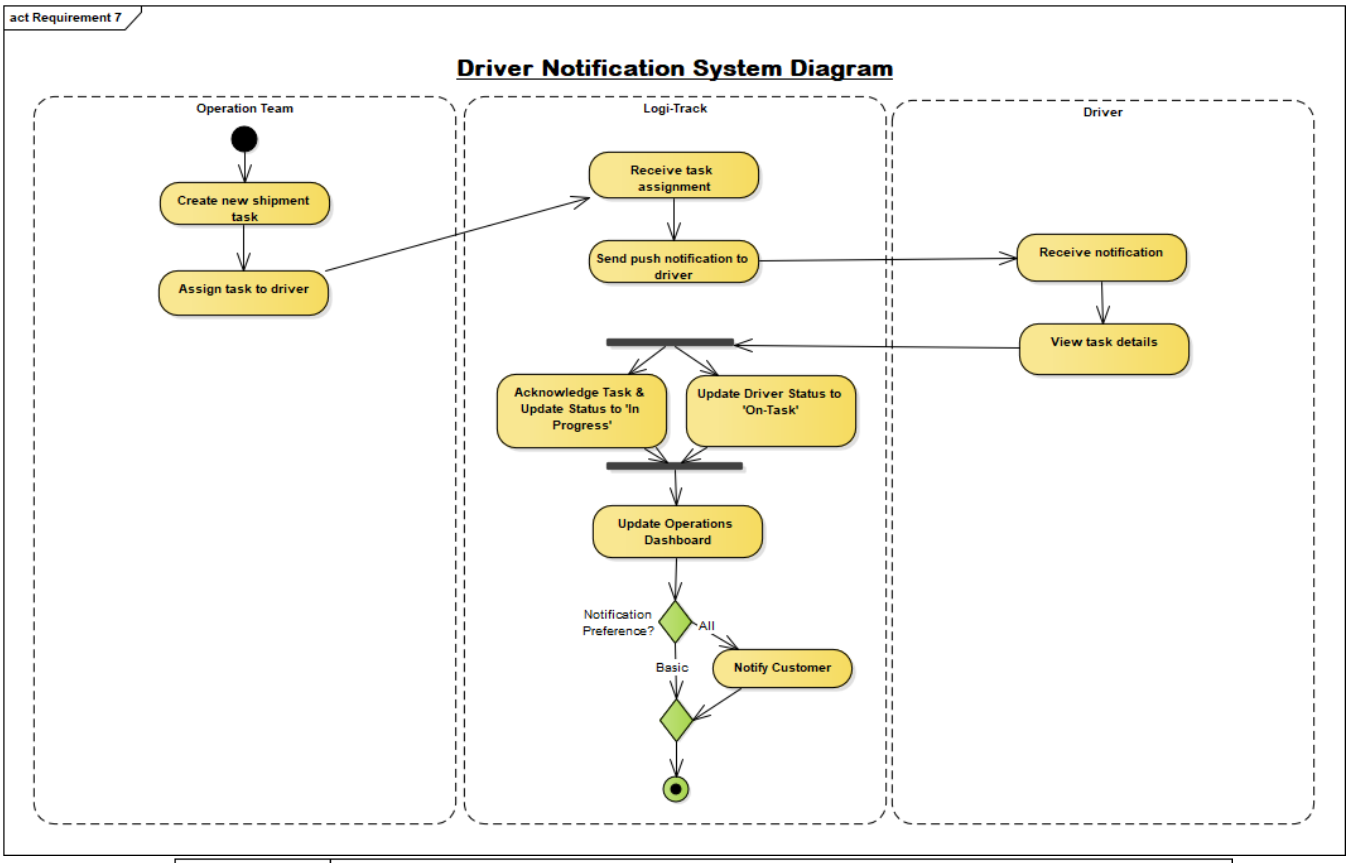


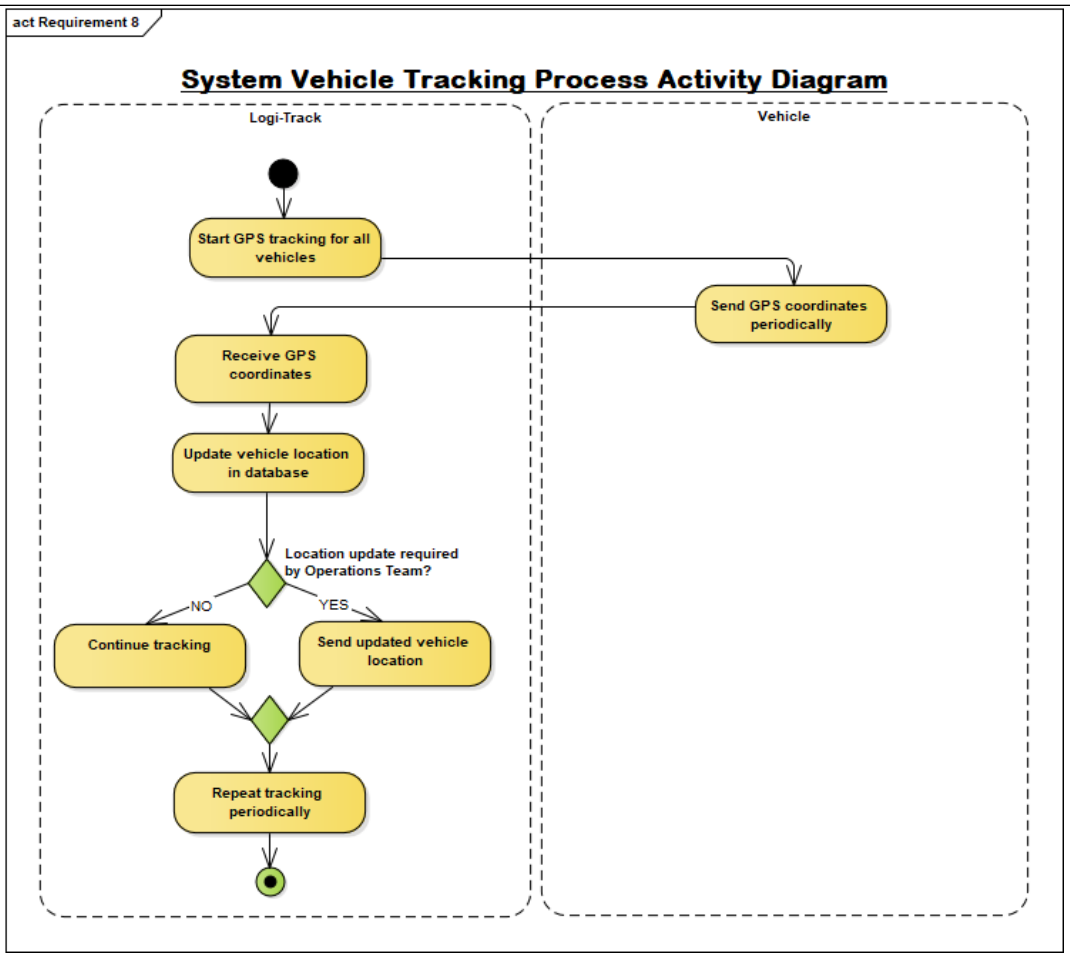

act Requirements 3
Vehicle Data Entry Flow
Logi-Track
Check if vehicle sensors are available
Sensors available?
YES
NO
Retrieve telemetry automatically
Request manual vehicle data
Store telemetry in system
Validate & store manual data
Proceed with monitoring / dispatch

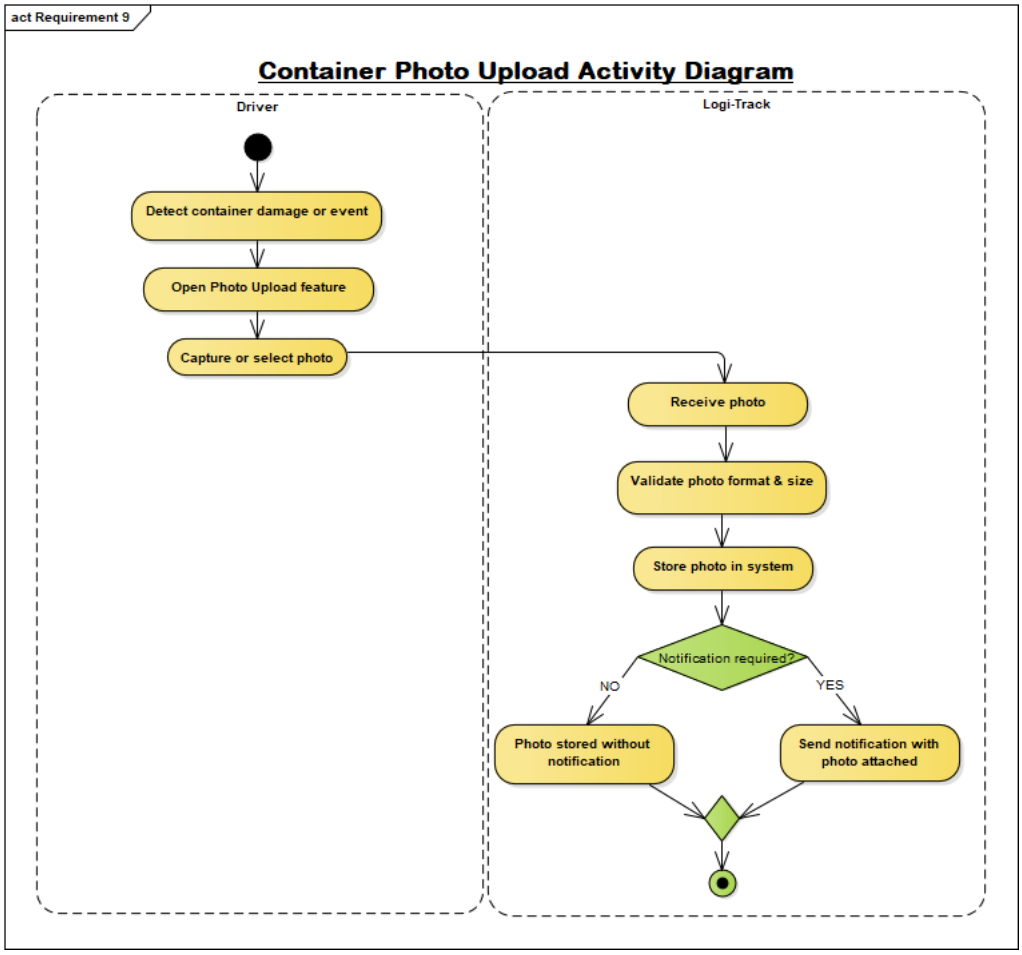

act Requirement 4
Shipment Status Monitoring & Notification (System)
Logi-Track
Monitor shipment status continuously
Status changed?
YES
NO
Determine type of status change
Continue monitoring
Generate notification message
Send notification via Web Dashboard
Send notification via Email
Send notification via Mobile App
Update customer status

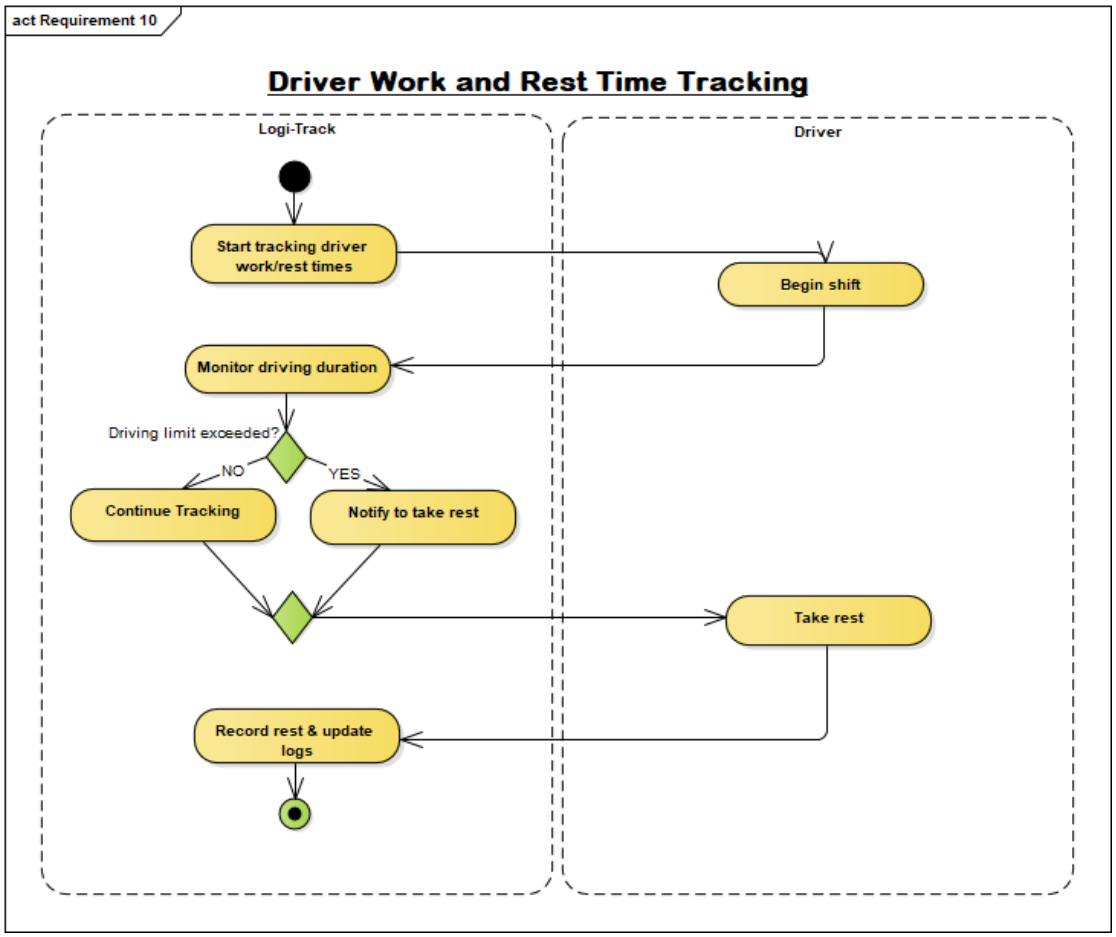

act Requirement 6
User Interaction with Tutorial Section Activity Diagram
User
Logi-Track
Open Application
Navigate to Tutorial Section
Load tutorial content
Display tutorial to user
Read tutorial content
User wants to continue?
YES
NO
Proceed to next tutorial step
Exit tutorial
Load next tutorial content
Repeat until tutorial complete

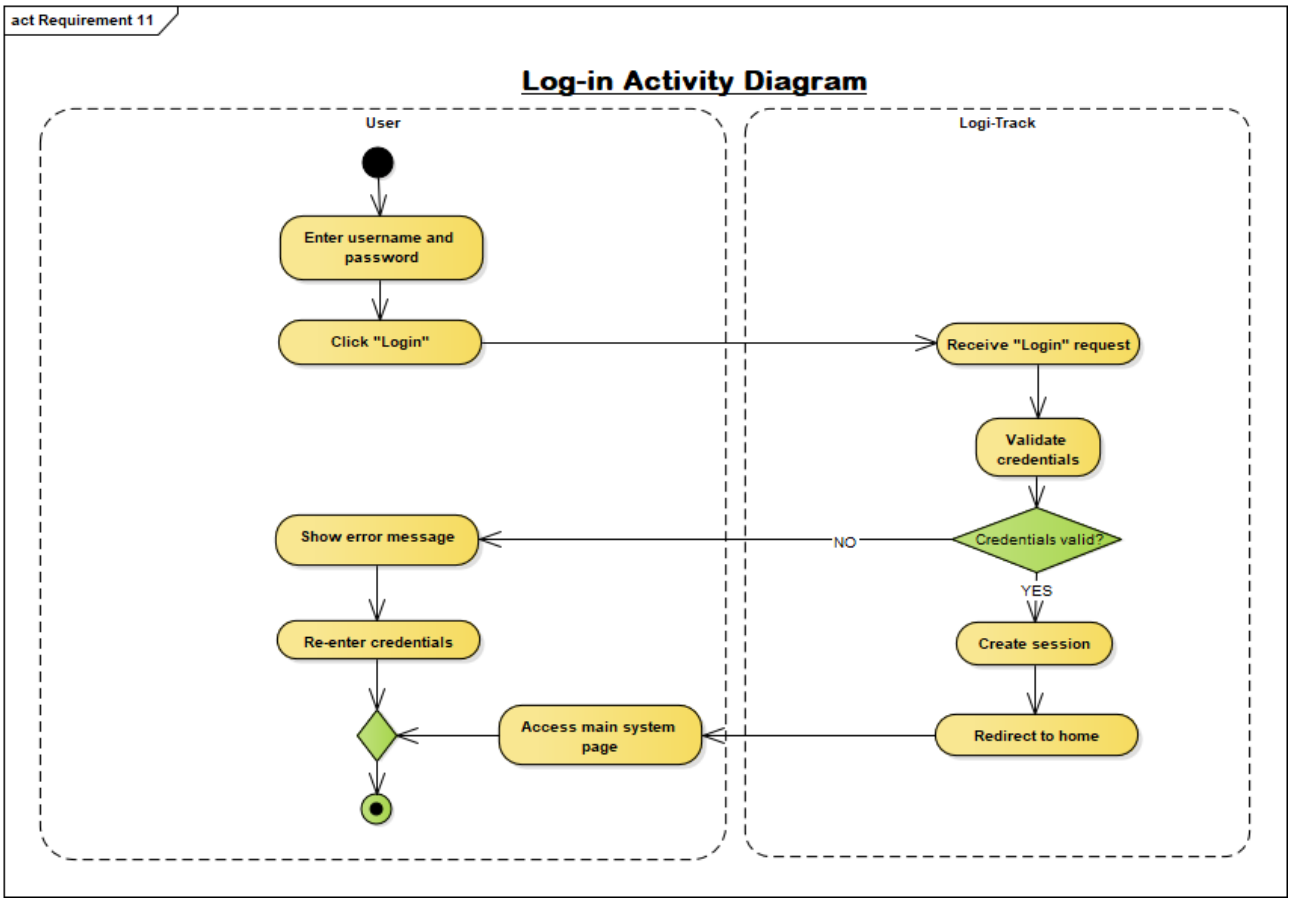

act Requirement 5
Invoice Creation & Finance Notification Flow
Driver
Logi-Track
Finance Department
Confirm delivery
Receive delivery confirmation
Create invoice record for the shipment
Validate invoice details
Send notification about new invoice
Receive invoice notification
Review invoice for processing

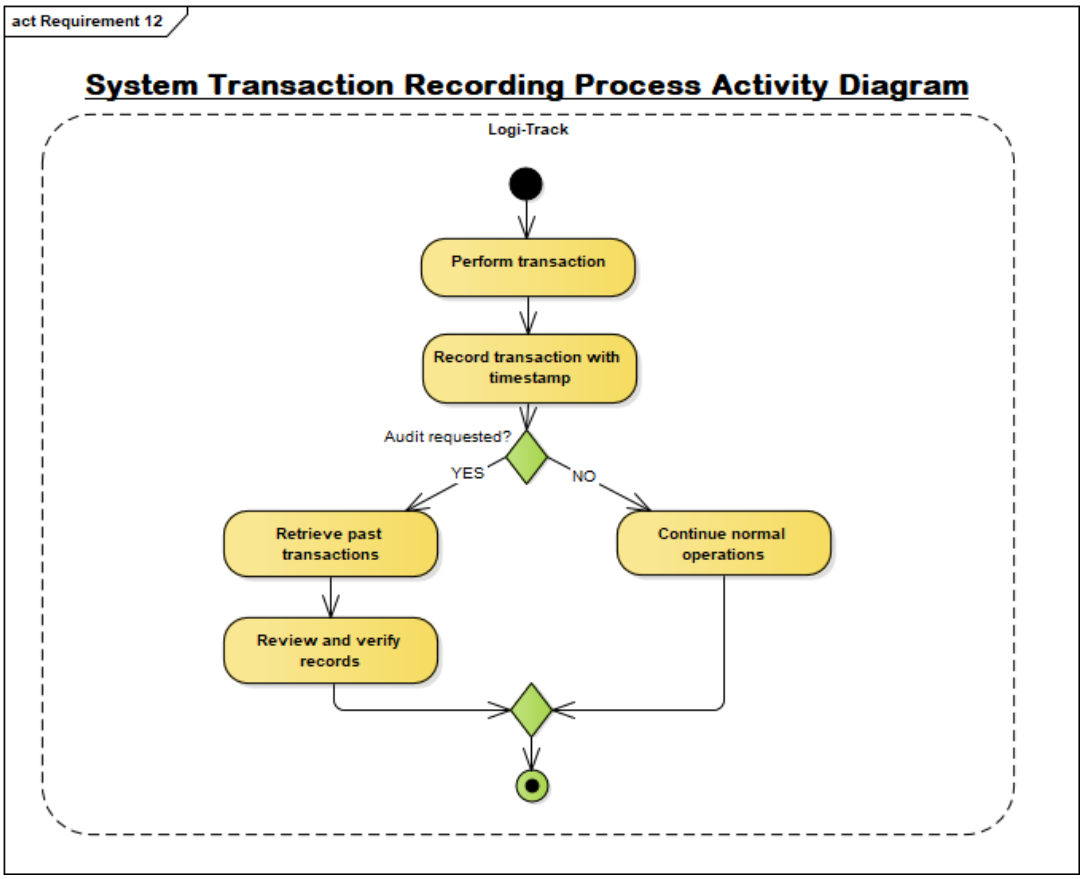

act Requirement 7
Driver Notification System Diagram
Operation Team
Logi-Track
Driver
Create new shipment task
Assign task to driver
Receive task assignment
Send push notification to driver
Receive notification
View task details
Acknowledge Task & Update Status to 'In Progress'
Update Driver Status to 'On-Task'
Update Operations Dashboard
Notification Preference?
All
Basic
Notify Customer


act Requirement 8
System Vehicle Tracking Process Activity Diagram
Logi-Track
Vehicle
Start GPS tracking for all vehicles
Send GPS coordinates periodically
Receive GPS coordinates
Update vehicle location in database
Location update required by Operations Team?
NO
YES
Continue tracking
Send updated vehicle location
Repeat tracking periodically

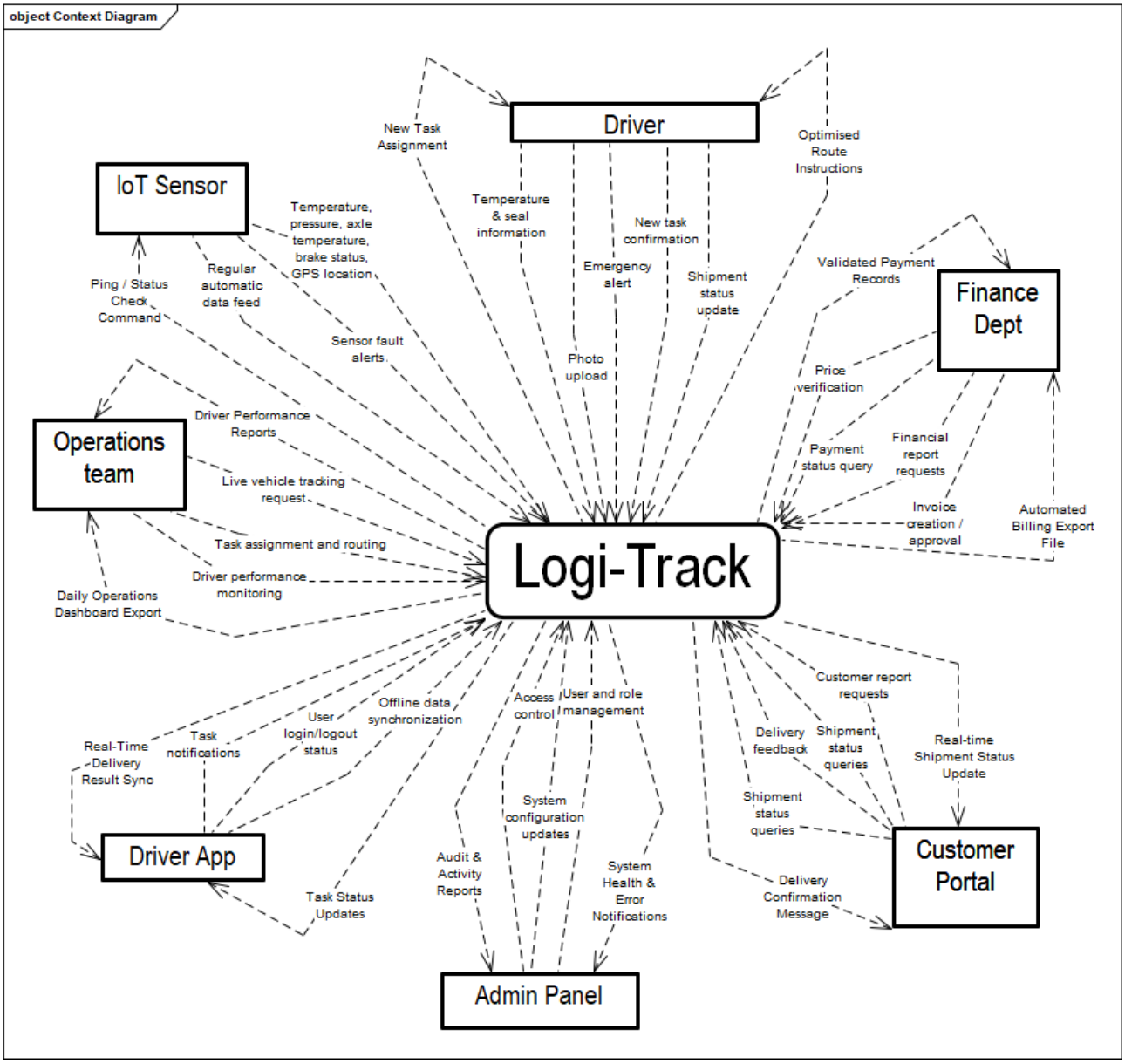

act Requirement 9
Container Photo Upload Activity Diagram
Driver
Logi-Track
Detect container damage or event
Open Photo Upload feature
Capture or select photo
Receive photo
Validate photo format & size
Store photo in system
Notification required?
NO
YES
Photo stored without notification
Send notification with photo attached


act Requirement 10
Driver Work and Rest Time Tracking
Logi-Track
Driver
Start tracking driver work/rest times
Begin shift
Monitor driving duration
Driving limit exceeded?
NO
YES
Continue Tracking
Notify to take rest
Take rest
Record rest & update logs

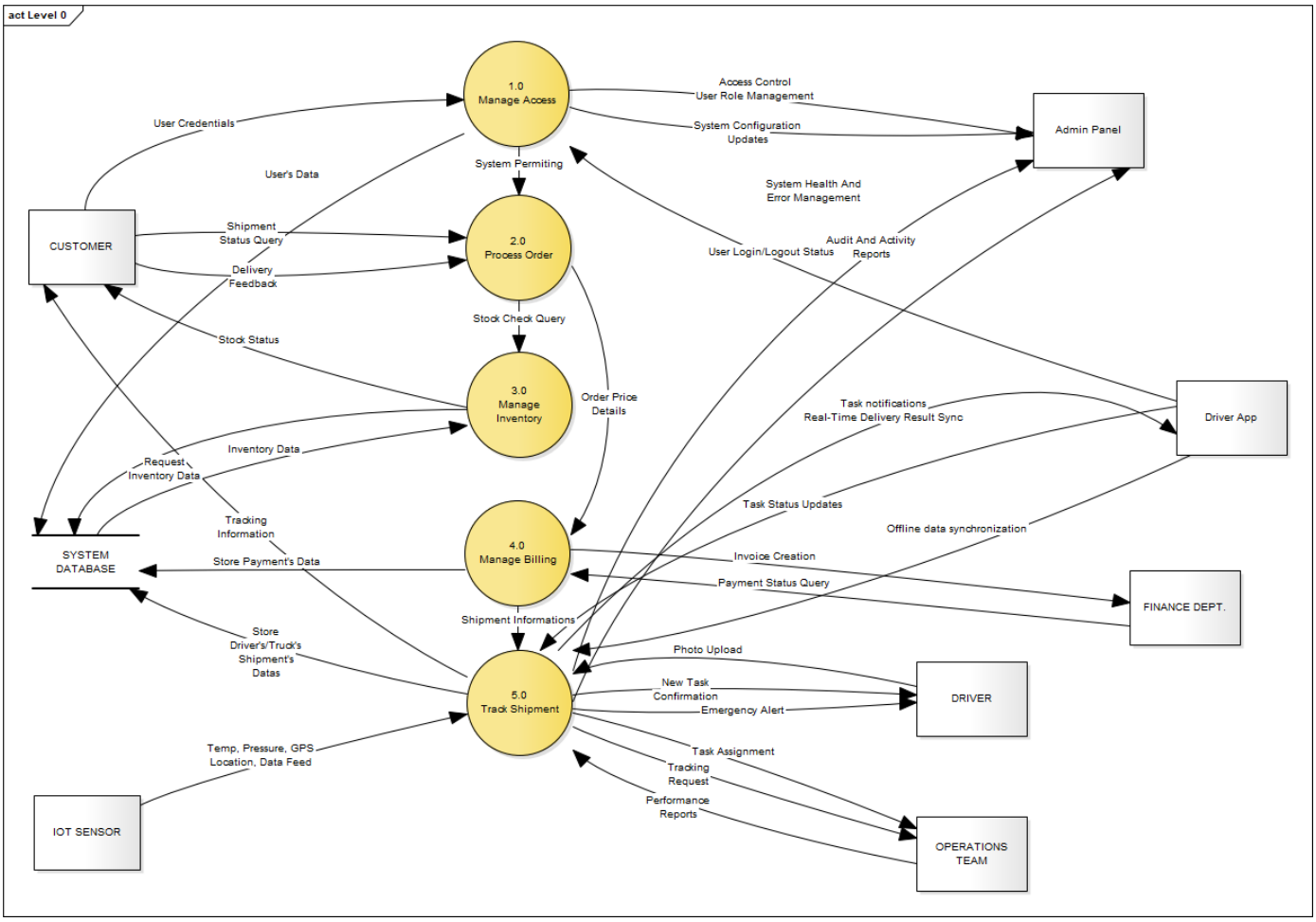

act Requirement 11
Log-in Activity Diagram
User
Logi-Track
Enter username and password
Click "Login"
Receive "Login" request
Validate credentials
Credentials valid?
NO
YES
Show error message
Re-enter credentials
Create session
Access main system page
Redirect to home

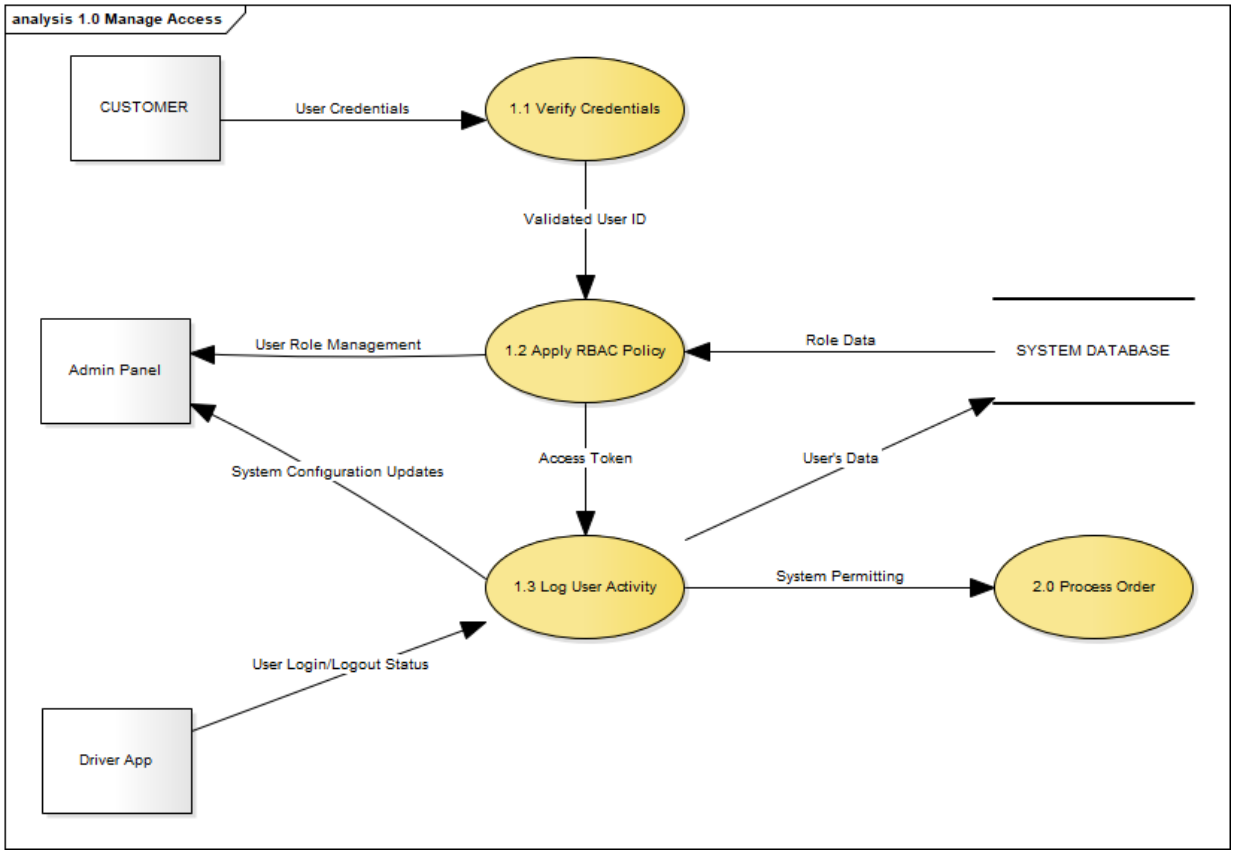

act Requirement 12
System Transaction Recording Process Activity Diagram
Logi-Track
Perform transaction
Record transaction with timestamp
Audit requested?
YES
NO
Retrieve past transactions
Continue normal operations
Review and verify records

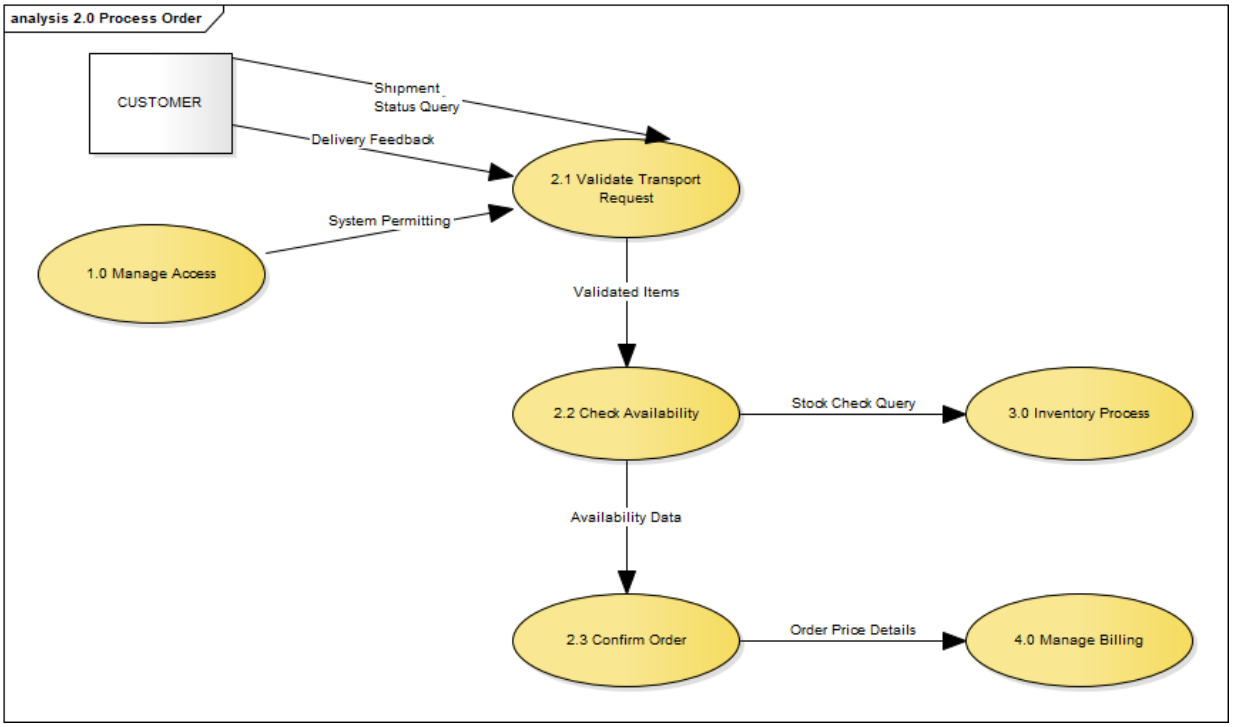
3.3. Context Model

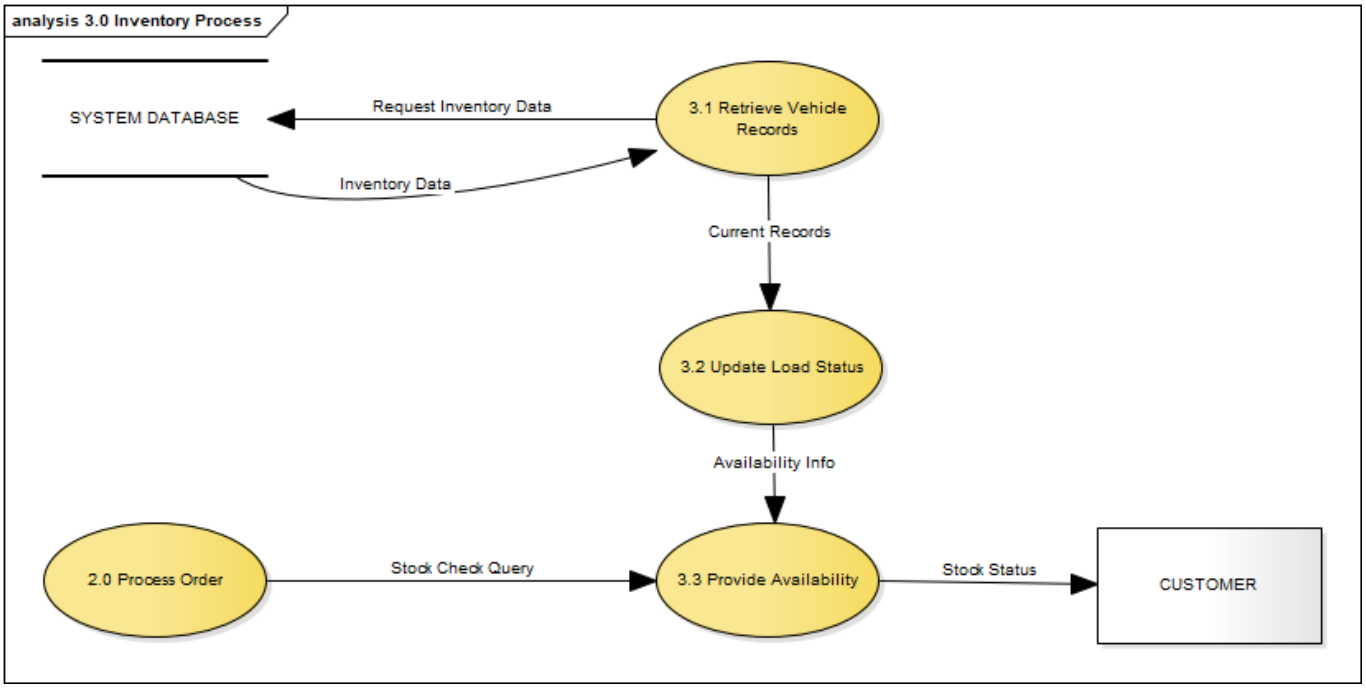


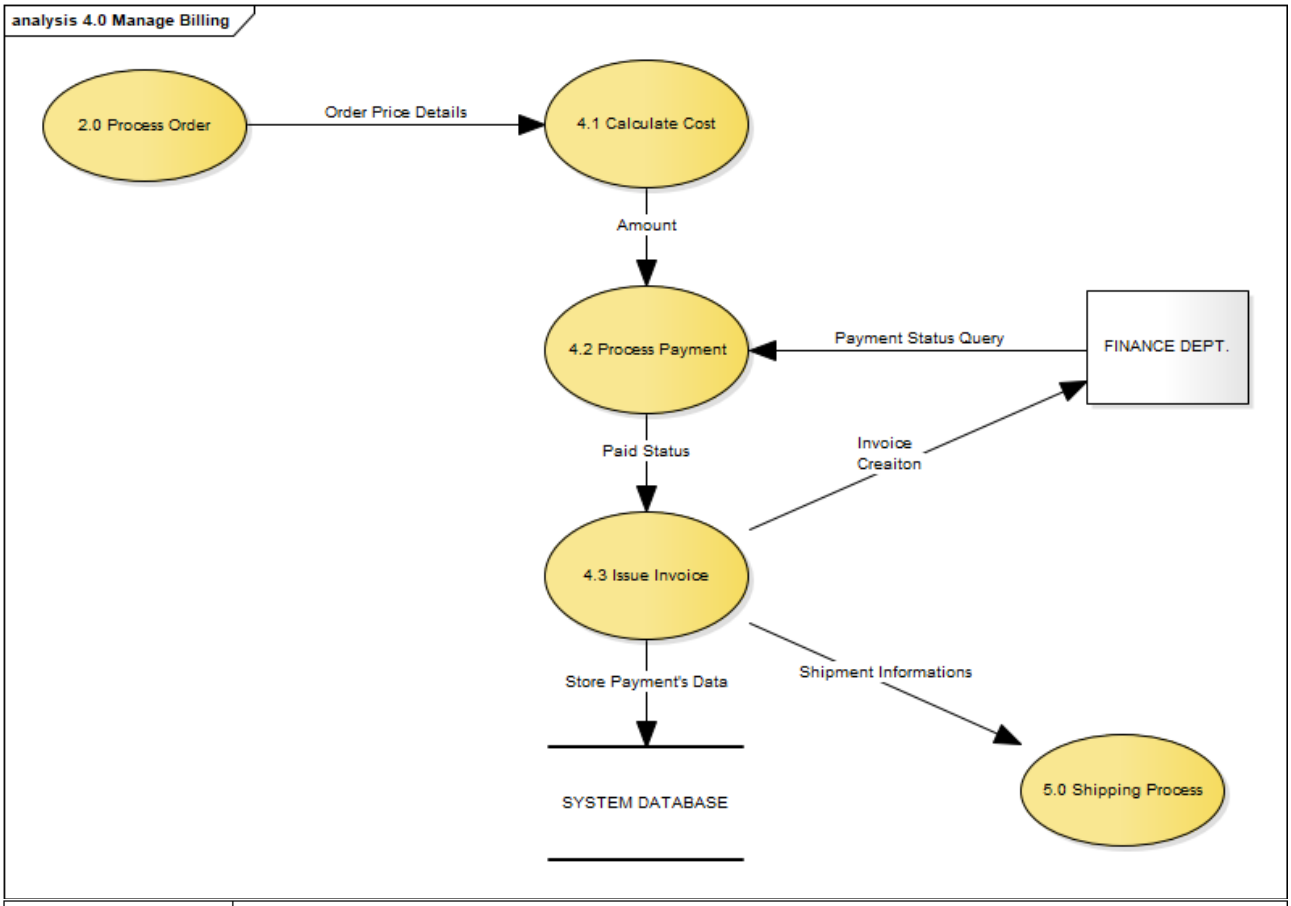
3.4. Data Flow Diagrams:

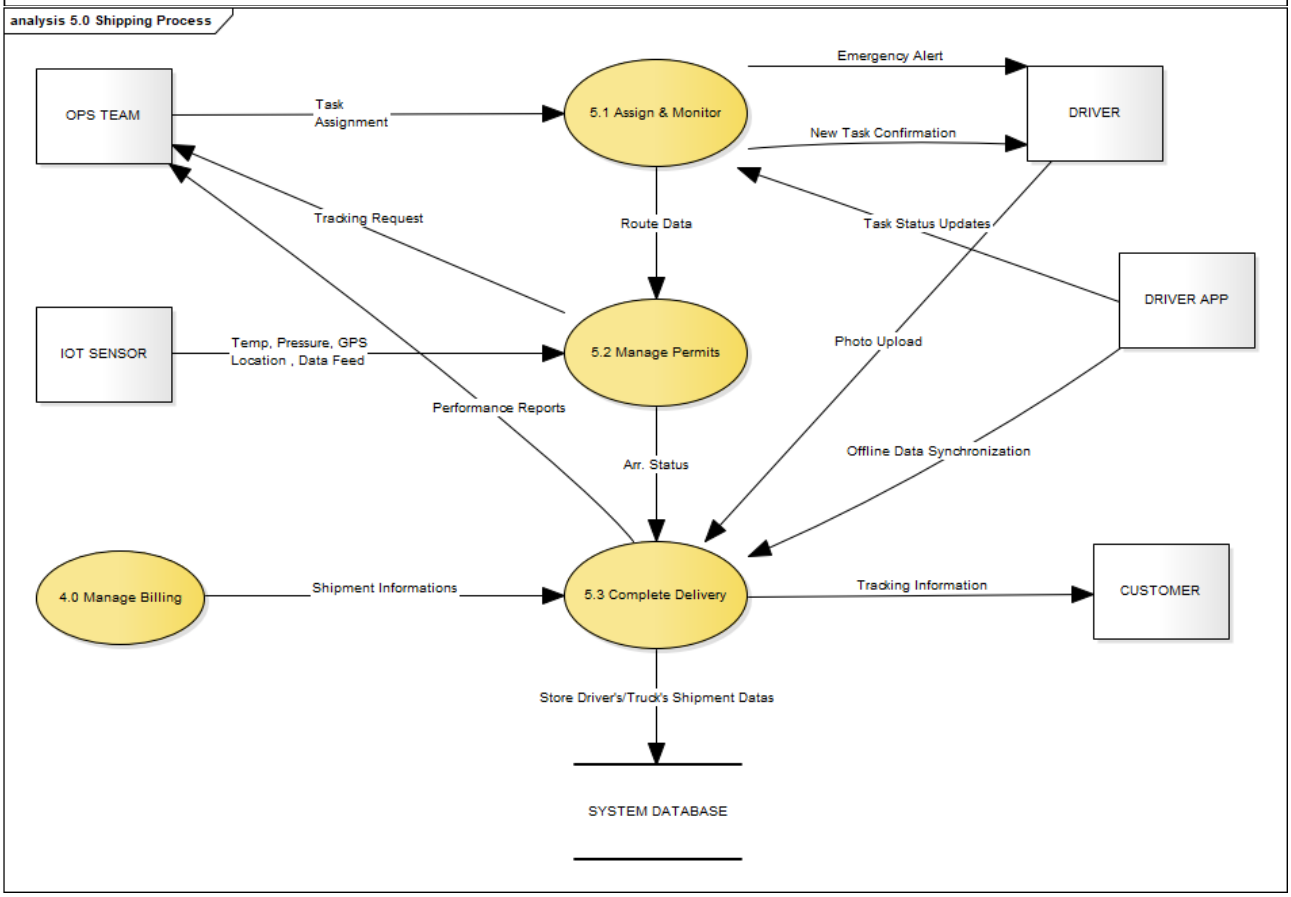




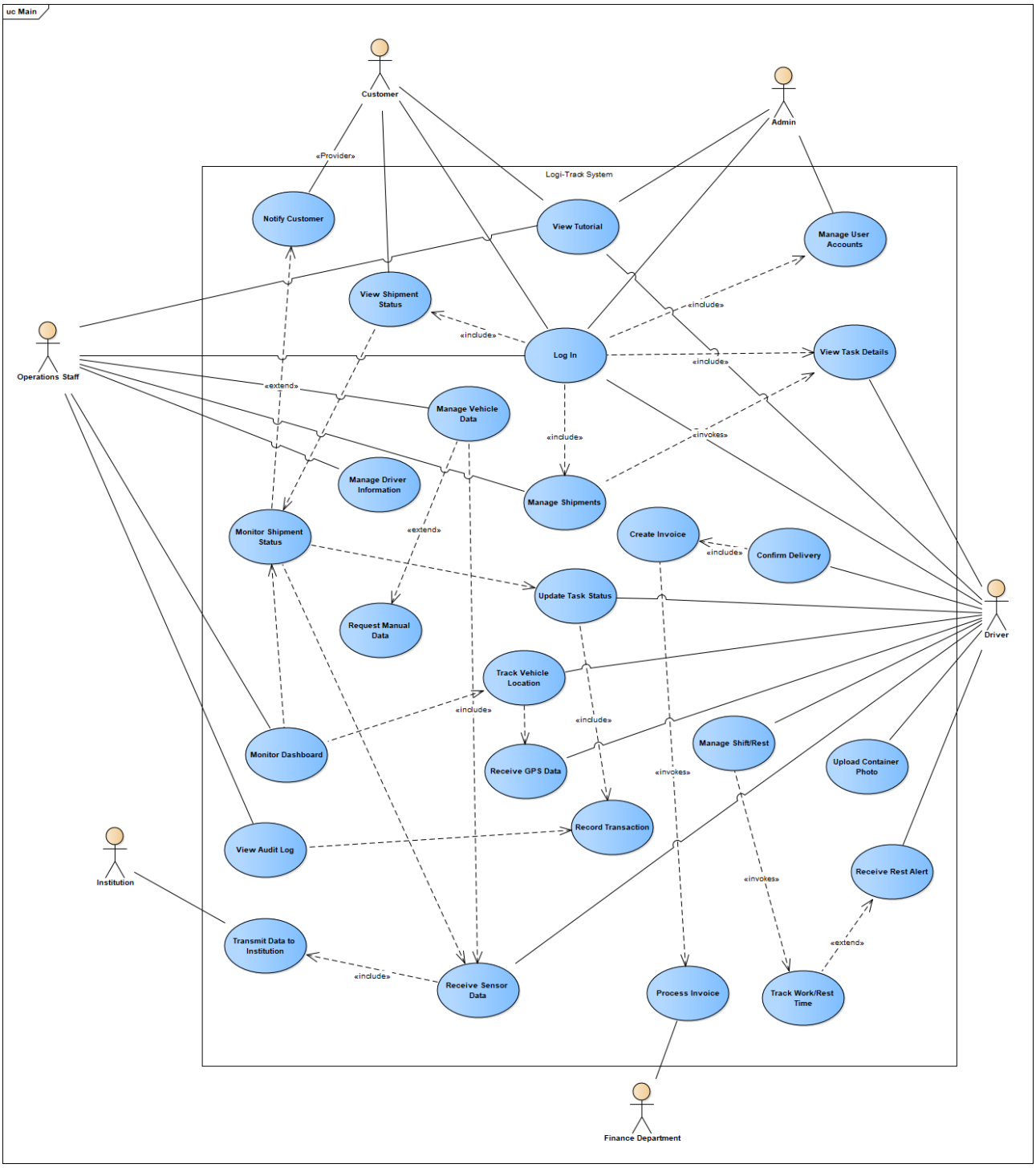

analysis 2.0 Process Order
CUSTOMER
Shipment Status Query
Delivery Feedback
2.1 Validate Transport Request
System Permitting
1.0 Manage Access
Validated Items
2.2 Check Availability
Stock Check Query
3.0 Inventory Process
Availability Data
2.3 Confirm Order
Order Price Details
4.0 Manage Billing


analysis 3.0 Inventory Process
SYSTEM DATABASE
Request Inventory Data
3.1 Retrieve Vehicle Records
Inventory Data
Current Records
3.2 Update Load Status
Availability Info
2.0 Process Order
Stock Check Query
3.3 Provide Availability
Stock Status
CUSTOMER

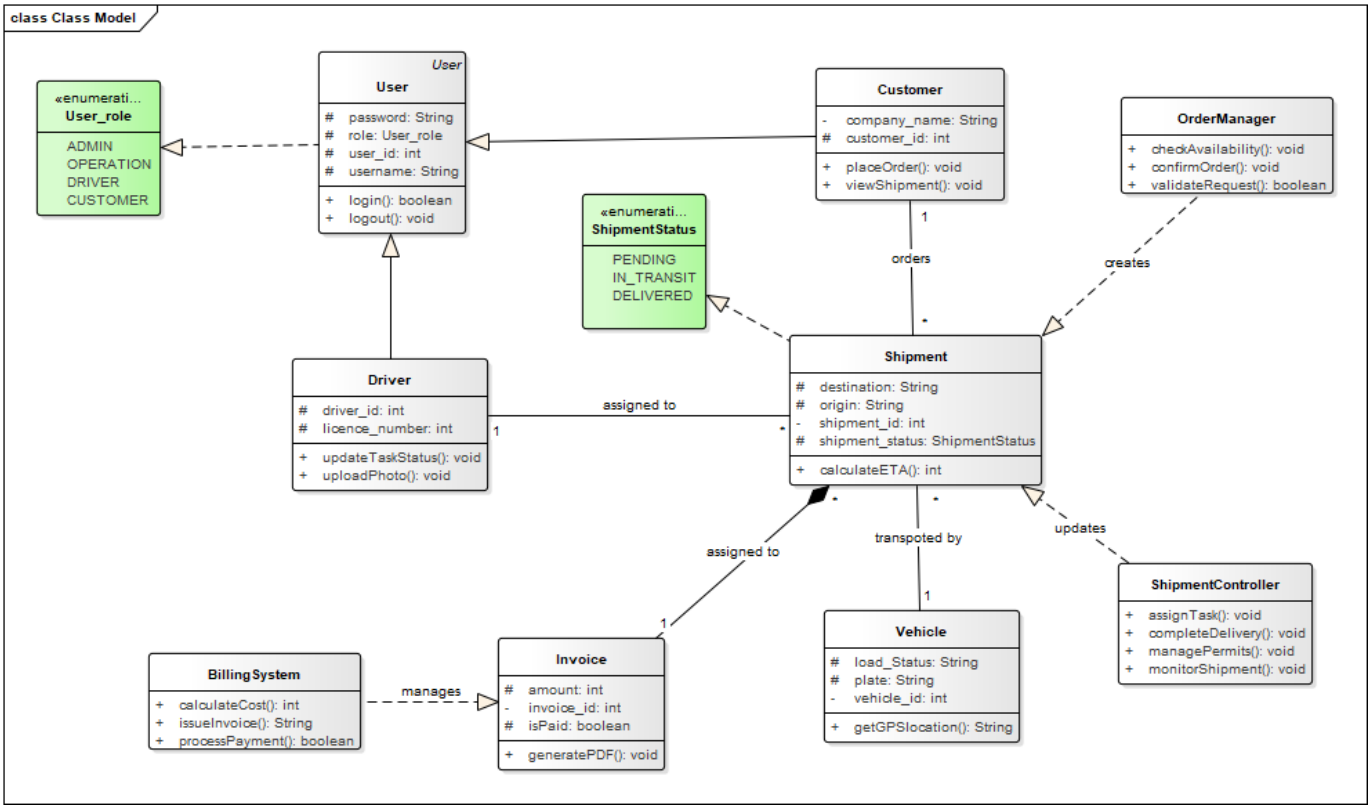

analysis 4.0 Manage Billing
2.0 Process Order
Order Price Details
4.1 Calculate Cost
Amount
4.2 Process Payment
Payment Status Query
FINANCE DEPT.
Paid Status
Invoice Creaiton
4.3 Issue Invoice
Store Payment's Data
Shipment Informations
SYSTEM DATABASE
5.0 Shipping Process


analysis 5.0 Shipping Process
OPS TEAM
Task Assignment
5.1 Assign & Monitor
Emergency Alert
New Task Confirmation
DRIVER
Tracking Request
Route Data
Task Status Updates
DRIVER APP
IOT SENSOR
Temp, Pressure, GPS Location , Data Feed
5.2 Manage Permits
Photo Upload
Performance Reports
Arr. Status
Offline Data Synchronization
4.0 Manage Billing
Shipment Informations
5.3 Complete Delivery
Tracking Information
CUSTOMER
Store Driver's/Truck's Shipment Datas
SYSTEM DATABASE

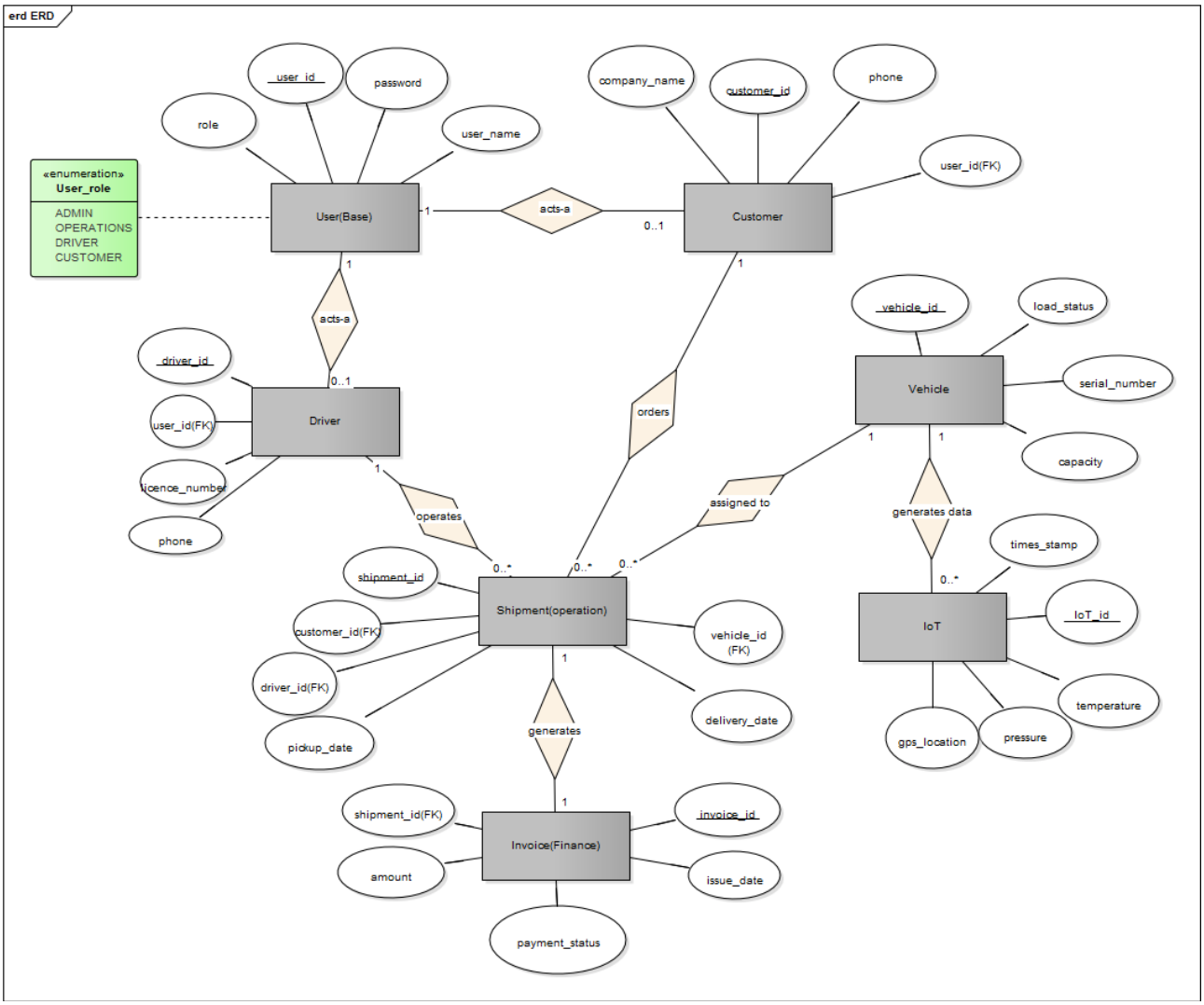
3.5. Use Case Diagram:



3.6. Class Diagrams:

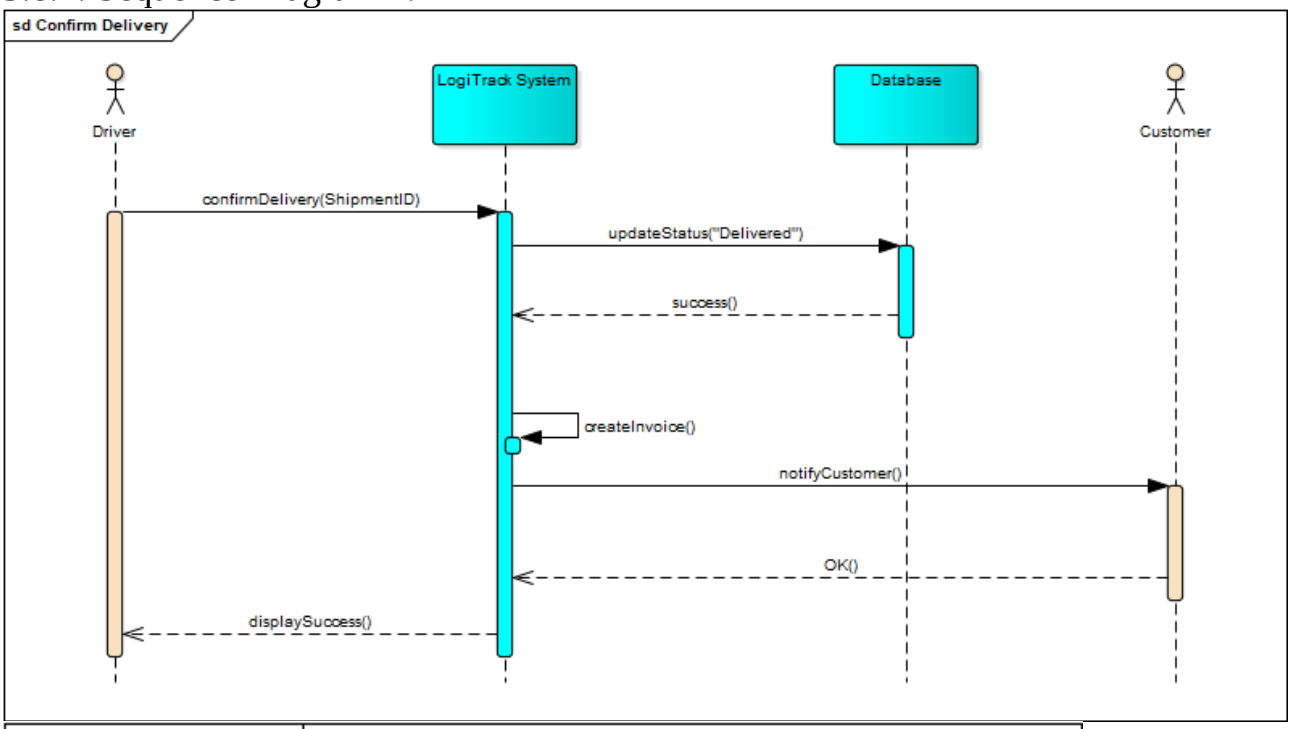


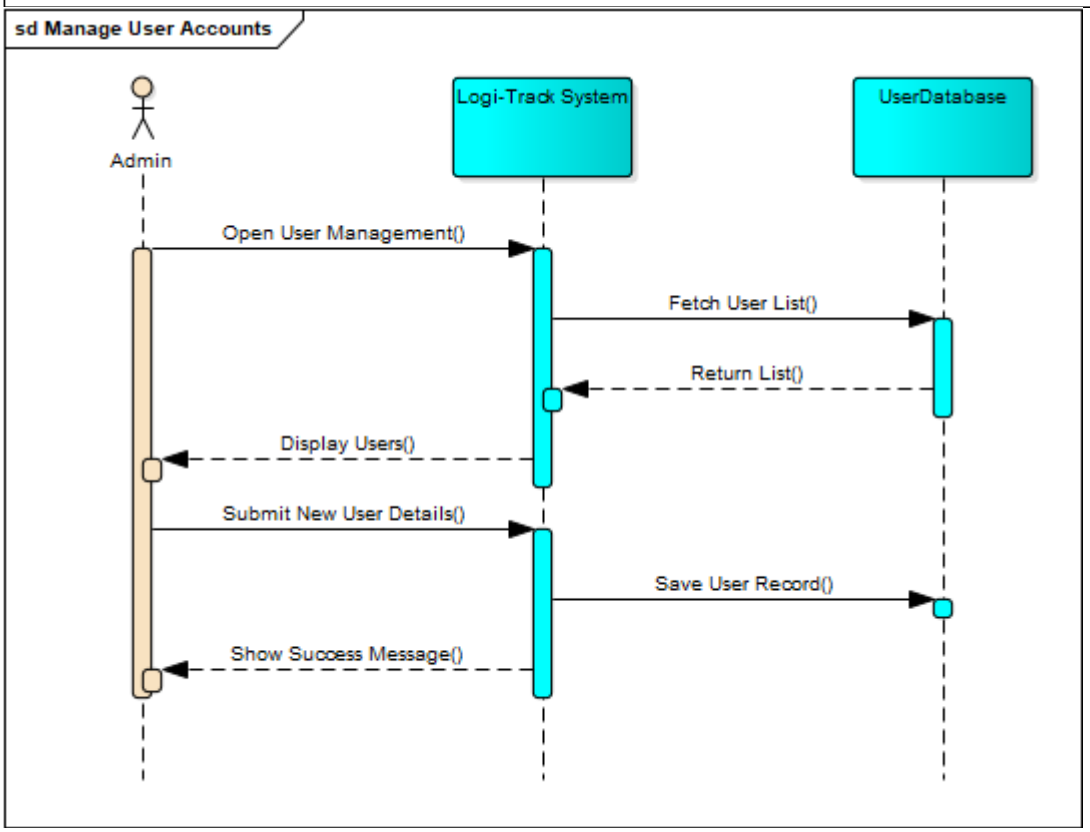
3.7. E/R Diagram:

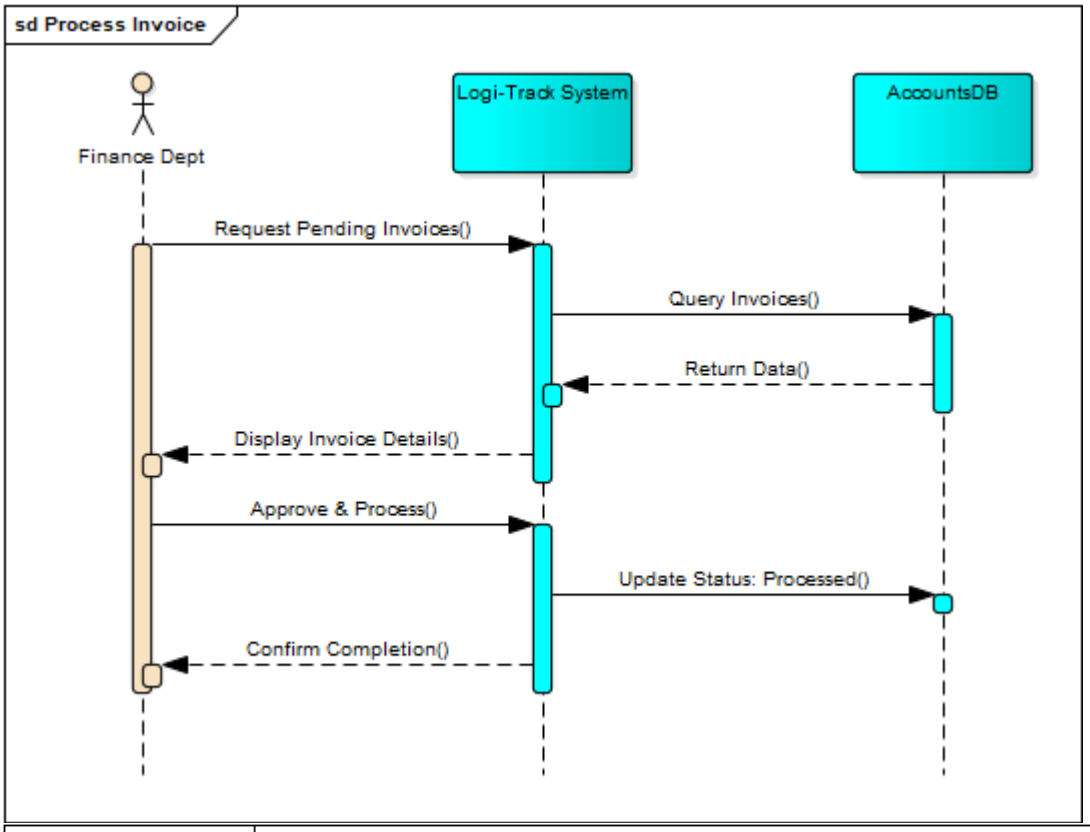


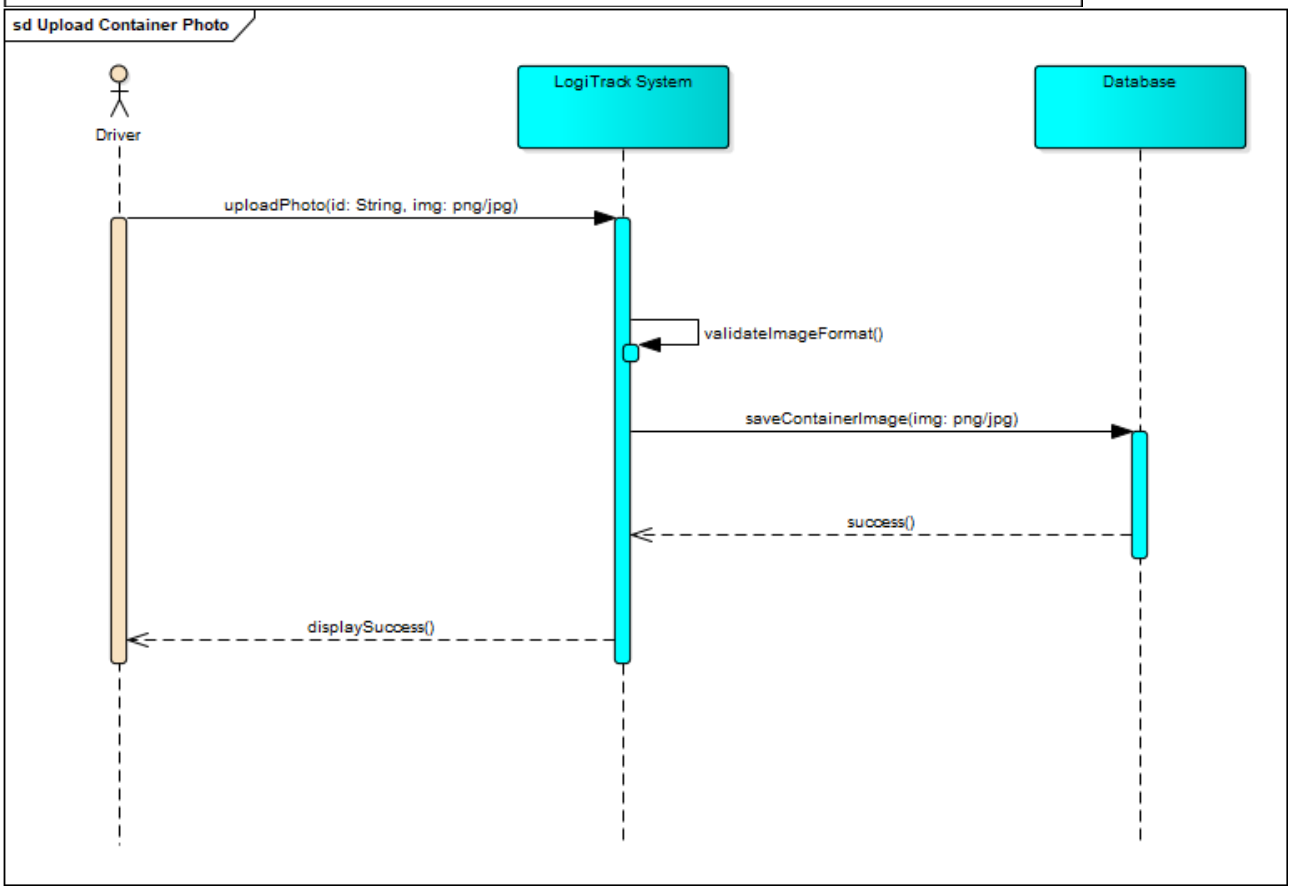
3.8. Sequence Diagrams

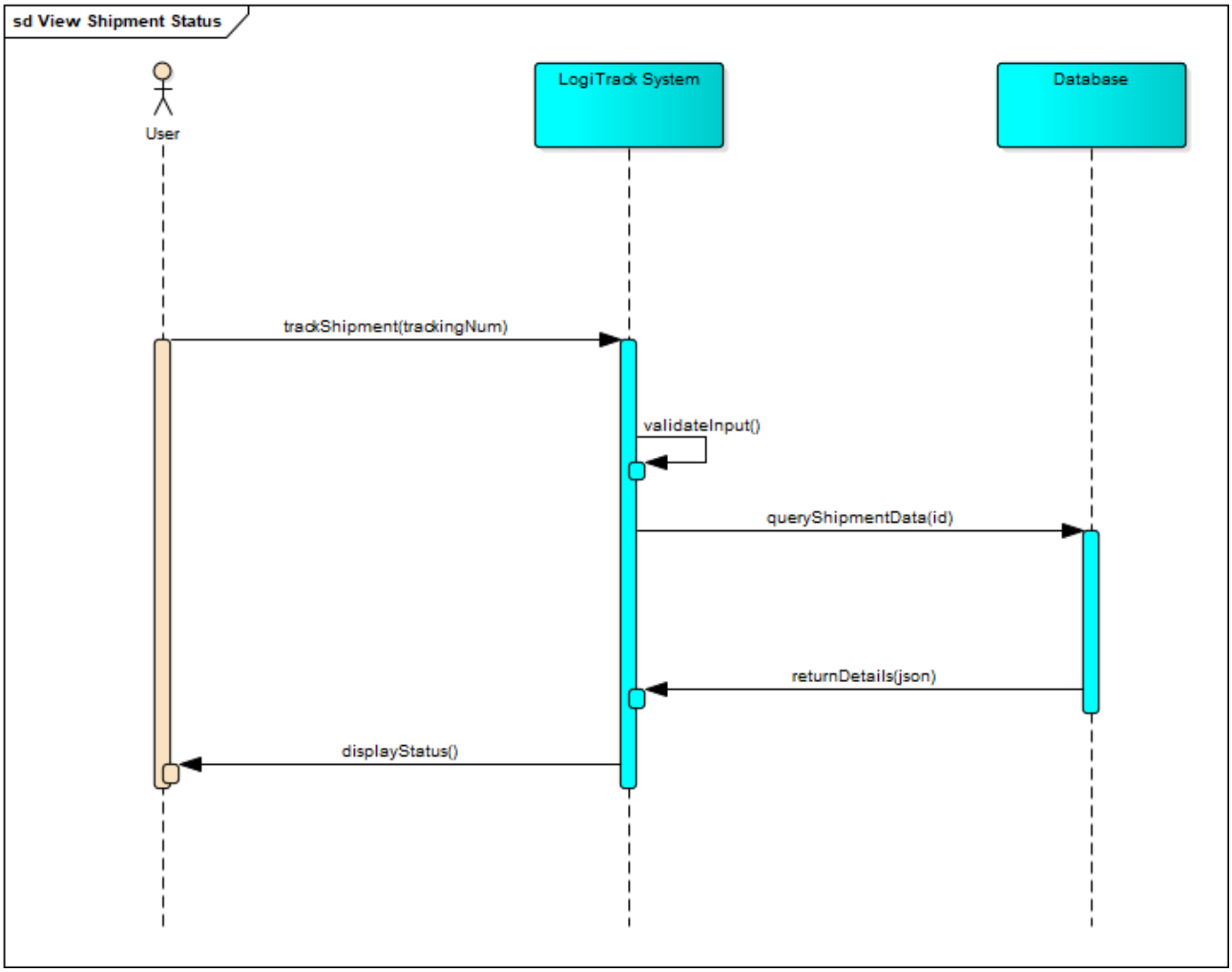
3.8.1. Sequence Diagram 1:





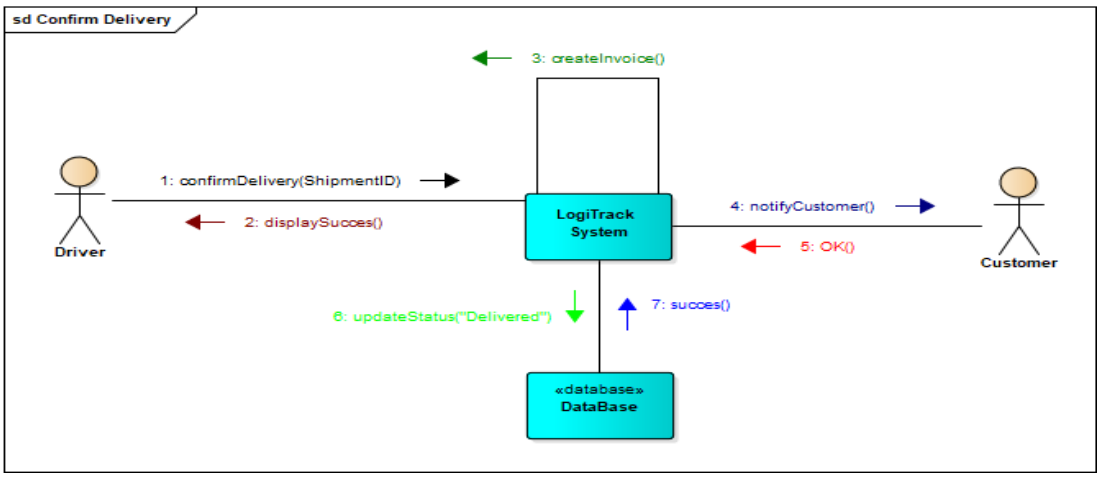

sd Process Invoice
Finance Dept
Logi-Track System
AccountsDB
Request Pending Invoices()
Query Invoices()
Return Data()
Display Invoice Details()
Approve & Process()
Update Status: Processed()
Confirm Completion()

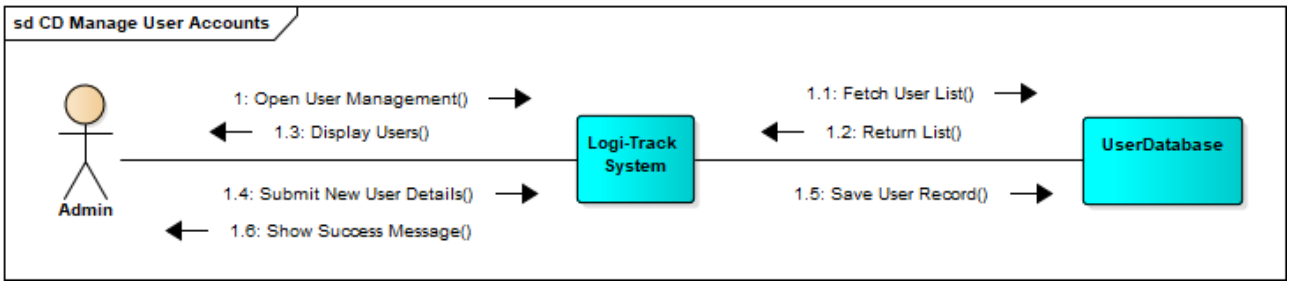

sd Upload Container Photo
Driver
LogiTrack System
Database
uploadPhoto(id: String, img: png/jpg)
validateImageFormat()
saveContainerImage(img: png/jpg)
success()
displaySuccess()

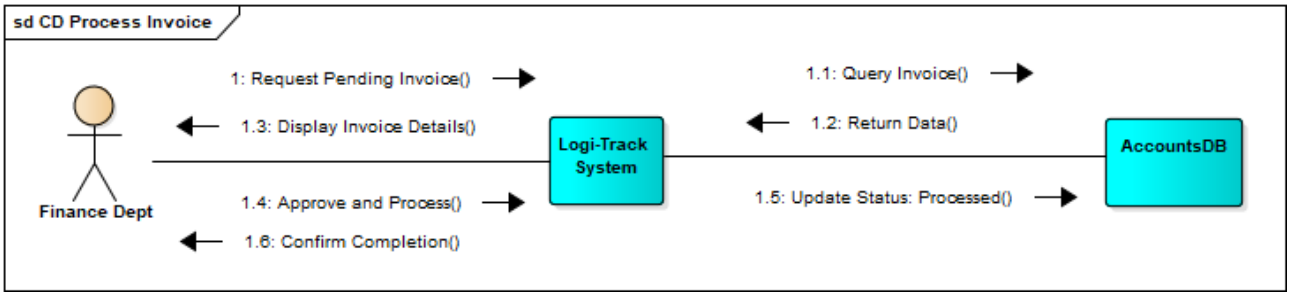


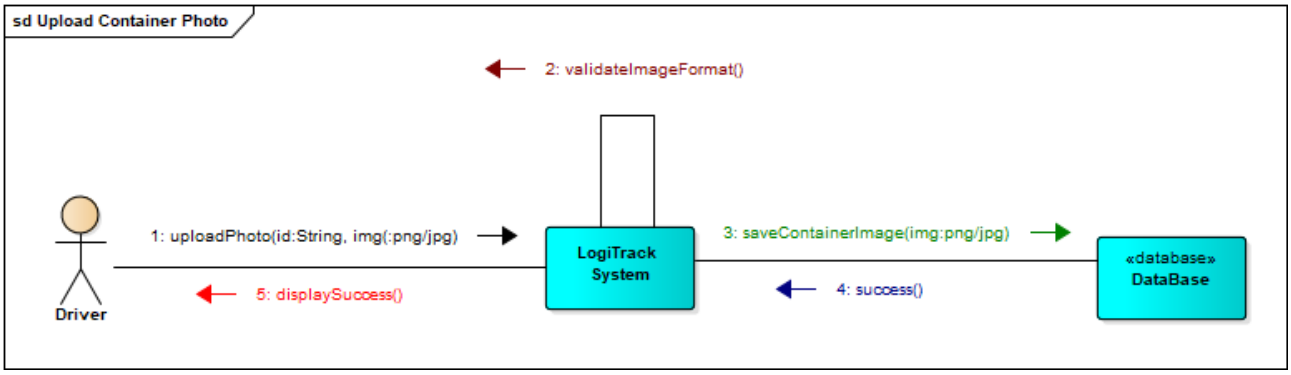
3.9. Communication Diagrams

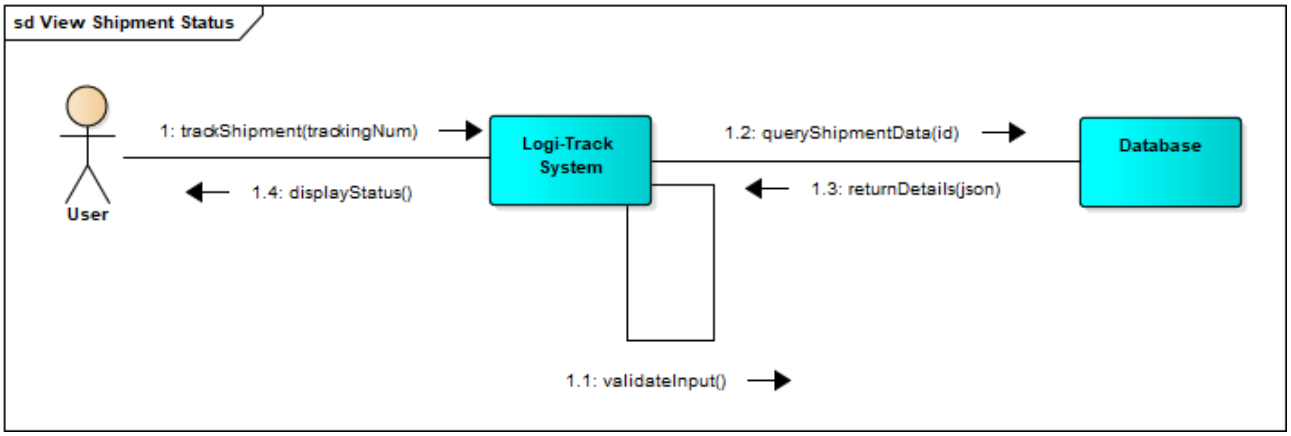
3.9.1. Communication Diagram 1:




sd CD Manage User Accounts
Admin
1: Open User Management()
1.3: Display Users()
1.4: Submit New User Details()
1.6: Show Success Message()
Logi-Track System
1.1: Fetch User List()
1.2: Return List()
1.5: Save User Record()
UserDatabase


sd CD Process Invoice
Finance Dept
1: Request Pending Invoice()
1.3: Display Invoice Details()
1.4: Approve and Process()
1.6: Confirm Completion()
Logi-Track System
1.1: Query Invoice()
1.2: Return Data()
1.5: Update Status: Processed()
AccountsDB


sd Upload Container Photo
2: validateImageFormat()
Driver
1: uploadPhoto(id:String, img(:png/jpg)
5: displaySuccess()
LogiTrack System
3: saveContainerImage(img:png/jpg)
4: success()
«database» DataBase


sd View Shipment Status
User
1: trackShipment(trackingNum)
1.4: displayStatus()
Logi-Track System
1.2: queryShipmentData(id)
1.3: returnDetails(json)
Database
1.1: validateInput()

4. Conclusion:

The Logi-Track project provides a comprehensive solution to the long-standing inefficiencies of the manual logistics industry. By replacing fragmented, communication-heavy processes with a unified, data-driven platform, the system successfully bridges the gap between operations teams, drivers, and customers.

The integration of real-time sensor monitoring, automated task notifications, and a transparent customer portal ensures that the movement of sensitive cargo is no longer a "black box" but a visible and manageable process. Furthermore, the implementation of Role-Based Access Control (RBAC) and automated compliance features like digital invoicing and timestamped auditing guarantees both data security and operational reliability.

In conclusion, Logi-Track is not just a tracking tool; it is a strategic asset that enhances operational efficiency, reduces human error, and fosters trust through transparency. By modernizing the workflow for perishable and temperature-sensitive goods, this project sets a new standard for logistics management, providing a scalable and future-proof foundation for companies to thrive in a digital-first global market.